Measurable tilings by abelian group actions

Terence Tao · 3 March 2022

https://terrytao.wordpress.com/2022/03/03/measurable-tilings-by-abelian-group-actions/

tao2typ · converted to Typst 2026-10-14 (UTC)

Jan Grebik, Rachel Greenfeld, Vaclav Rozhon and I have just uploaded to the arXiv our preprint "Measurable tilings by abelian group actions". This paper is related to an earlier paper of Rachel Greenfeld and myself concerning tilings of lattices $\mathbf{Z}^d$, but now we consider the more general situation of tiling a measure space X by a tile $A \subset X$ shifted by a finite subset F of shifts of an abelian group $G = (G, +)$ that acts in a measure-preserving (or at least quasi-measure-preserving) fashion on X. For instance, X could be a torus $\mathbf{T}^d = \mathbf{R}^d/\mathbf{Z}^d$, A could be a positive measure subset of that torus, and G could be the group $\mathbf{R}^d$, acting on X by translation.

If F is a finite subset of G with the property that the translates $f + A$, $f \in F$ of $A \subset X$ partition X up to null sets, we write $F \oplus A \underset{a.e.}{=} X$, and refer to this as a measurable tiling of X by A (with tiling set F). For instance, if X is the torus $\mathbf{T}^2$, we can create a measurable tiling with $A = [0, 1/2]^2 \mod \mathbf{Z}^2$ and $F = \{0, 1/2\}^2$. Our main results are the following:

- By modifying arguments from previous papers (including the one with Greenfeld mentioned above), we can establish the following "dilation lemma": a measurable tiling $F \oplus A \underset{a.e.}{=} X$ automatically implies further measurable tilings $rF \oplus A \underset{a.e.}{=} X$, whenever r is an integer coprime to all primes up to the cardinality $\#F$ of F.
- By averaging the above dilation lemma, we can also establish a "structure theorem" that decomposes the indicator function 1_A of A into components, each of which are invariant with respect to a certain shift in G. We can establish this theorem in the case of measure-preserving actions on probability spaces via the ergodic theorem, but one can also generalize to other settings by using the device of "measurable medial means" (which relates to the concept of a universally measurable set).
- By applying this structure theorem, we can show that all measurable tilings $F \oplus A = \mathbf{T}^1$ of the one-dimensional torus $\mathbf{T}^1$ are *rational*, in the sense that F lies in a coset of the rationals $\mathbf{Q} = \mathbf{Q}^1$. This answers a recent conjecture of Conley, Grebik, and Pikhurko; we also give an alternate proof of this conjecture using some previous results of Lagarias and Wang.
- For tilings $F \oplus A = \mathbf{T}^d$ of higher-dimensional tori, the tiling need not be rational. However, we can show that we can "slide" the tiling to be rational by giving each translate $f + A$ of A a "velocity" $v_f \in \mathbf{R}^d$, and for every time t, the translates $f + tv_f + A$ still form a partition of $\mathbf{T}^d$ modulo null sets, and at time $t = 1$ the tiling becomes rational. In particular, if a set A can tile a torus in an irrational fashion, then it must also be able to tile the torus in a rational fashion.
- In the two-dimensional case $d = 2$ one can arrange matters so that all the velocities v_f are parallel. If we furthermore assume that the tile A is connected, we can also show that the union of all the translates $f + A$ with a common velocity $v_f = v$ form a v-invariant subset of the torus.
- Finally, we show that tilings $F \oplus A = \mathbf{Z}^d \times G$ of a finitely generated discrete group $\mathbf{Z}^d \times G$, with G a finite group, cannot be constructed in a "local" fashion (we formalize this probabilistically using the notion of a "factor of iid process") unless the tile F is contained in a single coset of $\{0\} \times G$. (Nonabelian local tilings, for instance of the sphere by rotations, are of interest due to connections with the Banach-Tarski paradox; see the aforementioned paper of Conley, Grebik, and Pikhurko. Unfortunately, our methods seem to break down completely in the nonabelian case.)

Comments (9)

grpaseman · 3 March 2022

"my preprint"? I would expect "our preprint". (I see the joke now: How many mathematicians does it take to upload a preprint? I'll leave the punchline for you.)
[Corrected, thanks – T.]

> **Anonymous** · 4 March 2022
>
> He makes typos, relax

Anonymous · 6 March 2022

Is this the first time the expression HaHaHa has appeared in a maths paper 😉 ? Or is it an in-joke?

Anonymous · 6 March 2022

Is there any known example of similar tiling in the nonabelian case?

> **Terence Tao** · 7 March 2022
>
> We discuss some examples of nonabelian tilings in our paper. We know that the dilation lemma and structure theorem fail quite dramatically in these cases, and also have an example of a non-trivial local tiling in the nonabelian case. There are still some interesting questions concerning measurable tilings of higher-dimensional spheres by rotations (as studied in this previous paper), but unfortunately our methods do not have much to say about this situation.

Anonymous · 9 March 2022

Consider the (nonabelian) group of Mobius transformation on the Riemann sphere generated by translation $z \to z+1$ and inversion $z \to 1/z$. Is it possible to find a modular function corresponding to this group such that its fundamental domain creates a tilings of the Riemann sphere by the actions of this group?

> **Jas, the Physicist** · 10 March 2022
>
> I want to say yes. Did you think of an example?

Jas, the Physicist · 10 March 2022

What's a way of thinking of abelian group actions geometrically? When I think of a group action I naively only think of rotating S^1 in the complex plane which I think is abelian? But S^n has too perfect symmetry sometimes, so I can't imagine other things.

Jas, the Physicist · 10 March 2022

Forgive me for all the questions, especially if they are easily answered by understanding the paper (which I do not), but what is an example of a nonmeasurable tiling?
[A Vitali set gives a non-measurable tiling of the real line by the rationals -T.]